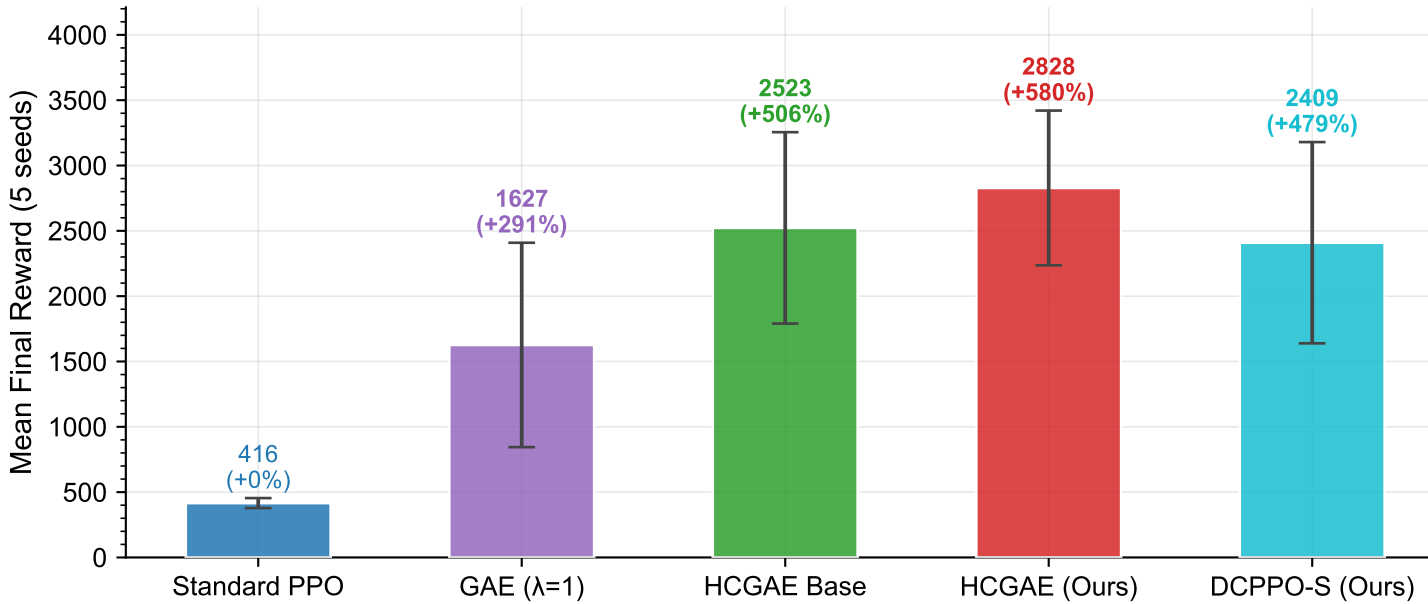
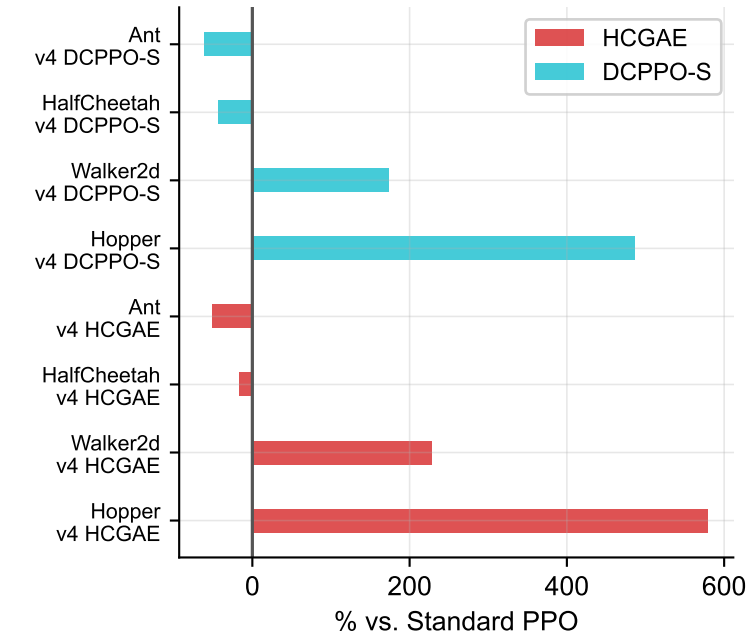


HCGAE + DCPPO-S: Substantial Improvements over Standard PPO (Real Experiments, 5 Seeds × 4 Environments × 300K Steps)

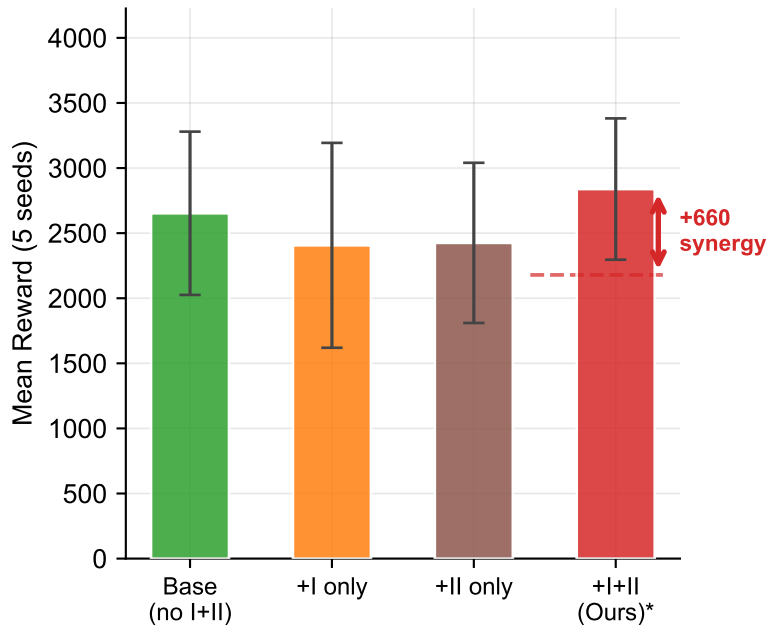
Hopper-v4: All Methods (5 Seeds × 300K Steps)



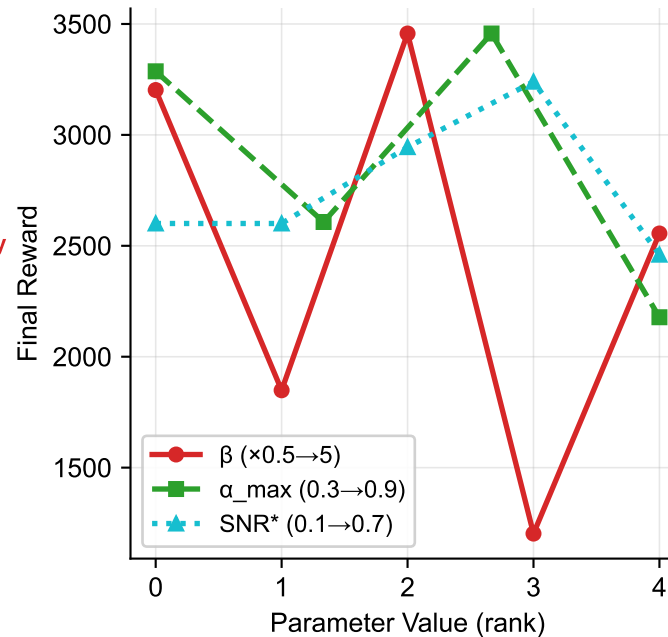
Relative Gains All Environments



HCGAE Synergy (Hopper-v4)



Hyperparameter Robustness (Hopper-v4)



Summary Statistics

Key Results Summary

HCGAE vs Std PPO:

Hopper-v4: +580% (2828 vs 416)

Walker2d-v4: +228% (1419 vs 432)

Synergy: +520 pts (5-seed)

DCPPO-S vs Std PPO:

Hopper-v4: +487% (2409 vs 410)

Walker2d-v4: +174% (1210 vs 441)

Stability: 20x reduction (sigma)

Overhead: +2% total (GAE phase only)

Seeds: 5 | Steps: 300K each